

/Users/halpern/teaching/352/labs/manual/master/rev_2023/alpha_2023.tex

Range of Alpha Particles in Air

1 Purpose

1. To study the energy loss of charged particles in matter. and in particular to measure the energy loss as a function of particle energy.

2 Theory

As charged particles traverse matter, they lose energy by ionizing atoms along their path. Once they have lost all their energy, they stop moving. At some level, this is not more complicated than how a ball rolls in sand. As the amount of energy lost per collision is calculable, each particle will have a path length or range in a given stopping material dependent on its incident energy. Since the important parameter is the number of collisions, the number of atoms passed is more important than the distance travelled. The relevant quantity is the **column density** which is a product of distance and density,

$$\text{Column Density} = \rho x \frac{\text{atoms}}{\text{cm}^2} \quad (1)$$

Thus for any particular type of particle the range is a definite function of the energy as shown in Figure 1. The mean range of particles is defined as shown in Figure 2, where n_o is the number of particles reaching a detector with no stopping material present and n is the number reaching the detector with stopping material of x cm and density ρ between the source and the detector.

The range distribution is broadened, as you see in figure 2 because the process we are dealing with is inherently random and because most particle sources have an intrinsic energy spread due to the emission process and also because of the thickness of the source itself causing a range of energy loss right at the start.

If we only want to know how far a particle travels, Figure 2 would be enough information. Suppose we are making smoke detectors. We want to know how far to put the detector from

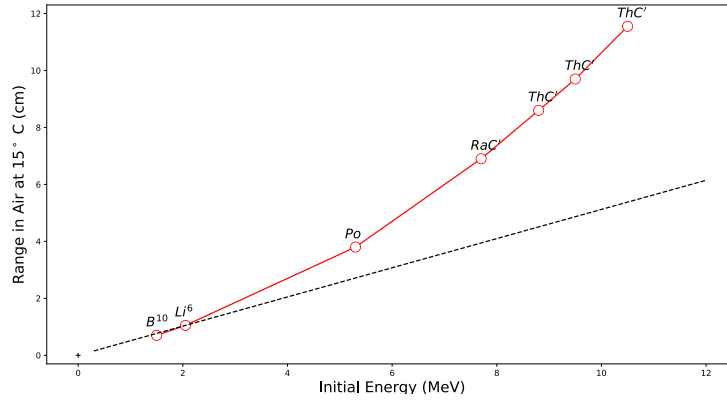


Figure 1: Range-energy relationship for α particles. See that higher energy α particles travel further. Notice that, to make sense, the density is specified.

the source so that the system is operating at the steep edge of the curve. Small changes in air density would result in large changes in the number of particles reaching the detector.

But we would like to explore the underlying mechanisms of energy loss so we will measure losses *as a function of particle energy*.

The rate of energy loss of alpha particles is given as

$$-\frac{dE}{dx} = \frac{4\pi Z^2}{m_e v^2} \left(\frac{e^2}{4\pi\epsilon_0} \right)^2 n_e \ln \frac{2m_e v^2}{I} \quad (2)$$

where I is some average ionization potential of the material, Z is the atomic number of the incoming particle, v is its (non-relativistic) velocity, and n_e is the electron density of the scattering material. This is plotted for Nitrogen gas in Figure 3. Notice that the stopping power drops at high particle energy with a rate between $1/\text{Energy}$ and $1/\text{Velocity}$.

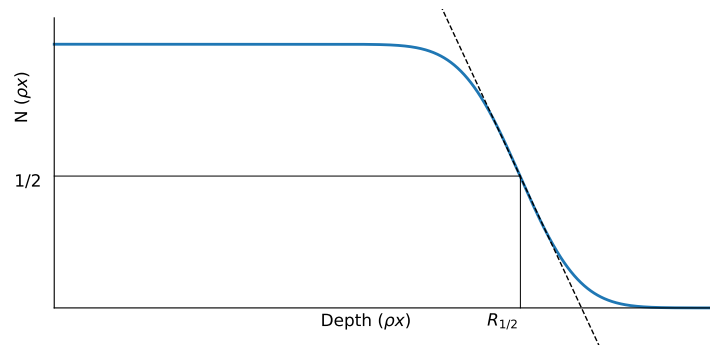


Figure 2: The extrapolated number-distance range R_n exceeds the mean range R by 0.886α , where α is the range-straggling parameter.

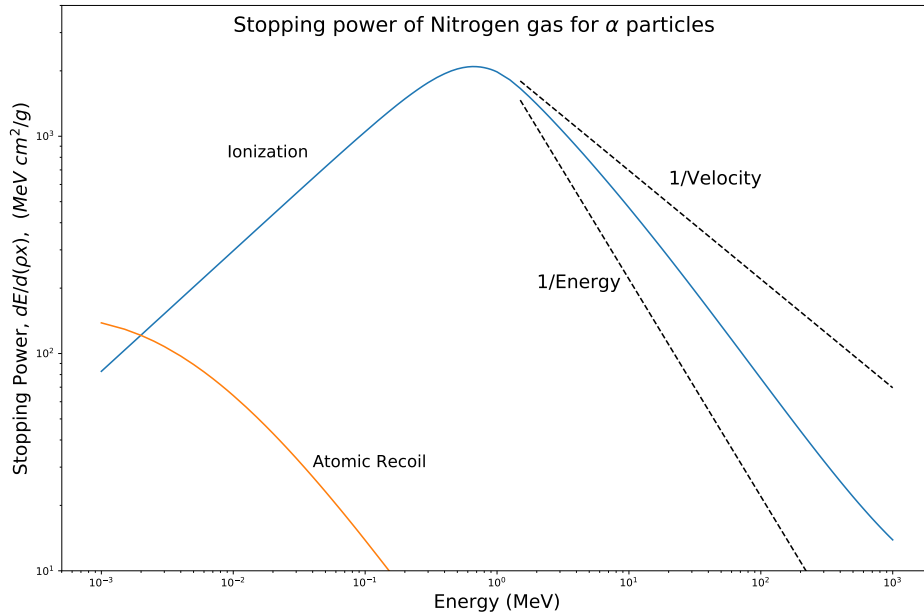


Figure 3: The stopping power of Nitrogen gas, from NIST (<https://physics.nist.gov/PhysRefData/Star/Text/ASTAR.html>). At low energies the α particle does not get near enough to the core of an atom to ionize it. At high energies it goes by too quickly for the electrical forces to have a large effect.

High energy particles do not lose much energy per collision. However, as they loose energy, they slow and effect of each interaction grows. Essentially, the change in momentum is proportional to force times time, and it takes a lot longer for a slow particle to move past a nucleus. The result is that charged particles deposit most of their energy right at the end of their travel. This is called a Bragg curve. A few examples are shown in Figure 4.

Measuring a Bragg curve for α particles in air is a main goal of this experiment.

While the concepts explored in this experiment have plenty of applications in nuclear physics and nuclear medicine, there are also some less obvious but still interesting applications. In particular, semiconductors are often doped by irradiating them with atomic nuclei and tuning the projectile energy to adjust the penetration depth of dopants into the material.

If you look up Bragg curves in the scientific literature you will find a lot of medical references. This is because, by adjusting the incident energy, clinicians can tune the depth of energy deposit in a material and target, for example, a tumor while somewhat sparing material around it.

(b) Bragg curves of O-15 ion beam

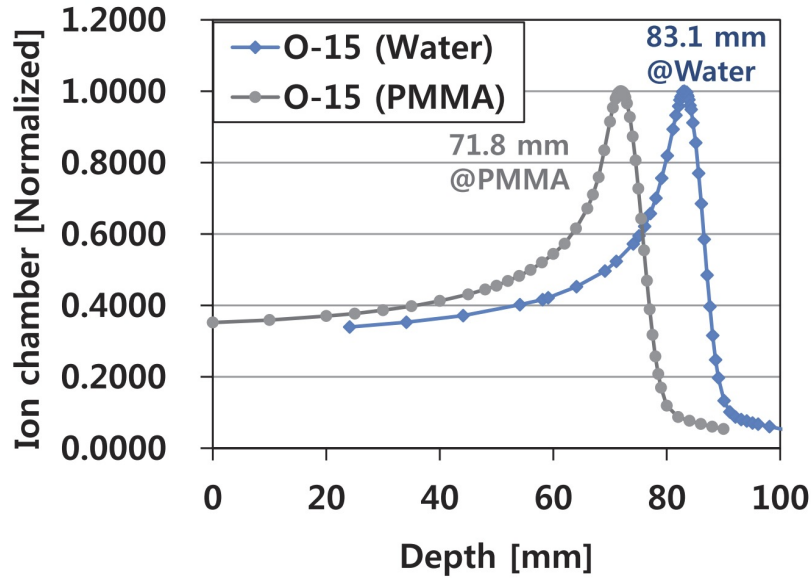


Figure 4: Range-energy relationship for α particles.

3 Procedure

The apparatus we will use is shown schematically in Figure 5. It consists of a radioactive Americium source and a detector in a steel can which can be evacuated. Using a Raspberry Pi and commands described in an Appendix you will be able to pump the air out of the can or to slowly allow air to flow back in.

Using the digital voltmeter, the Pi can read the signal from a pressure gauge, which you will calibrate by measuring the voltage reading at vacuum and at $P=1$ atmosphere.

Signals from the detector, as shown in Figure 6. You will measure these signals, both their shape and their rate, using commands from the Raspberry Pi to an oscilloscope.

The bias circuitry for the solid state detector should be on. If it is not, see the Appendix.

The trace in Figure 6 was collected by setting the oscilloscope to trigger on a rising edge. We recommend doing this adjustment at low pressures, near vacuum, because α particles are both plentiful and energetic in that configuration. The height of this signal is proportional to the particle energy. You can measure this by querying the oscilloscope for traces and analyzing the captured data in python. This is probably the most accurate way to proceed, but it is slow due to data transfer limits. Alternately, you can query the oscilloscope for the peak-to-peak amplitude or the *rms* variation of a trace. This approach is faster but less

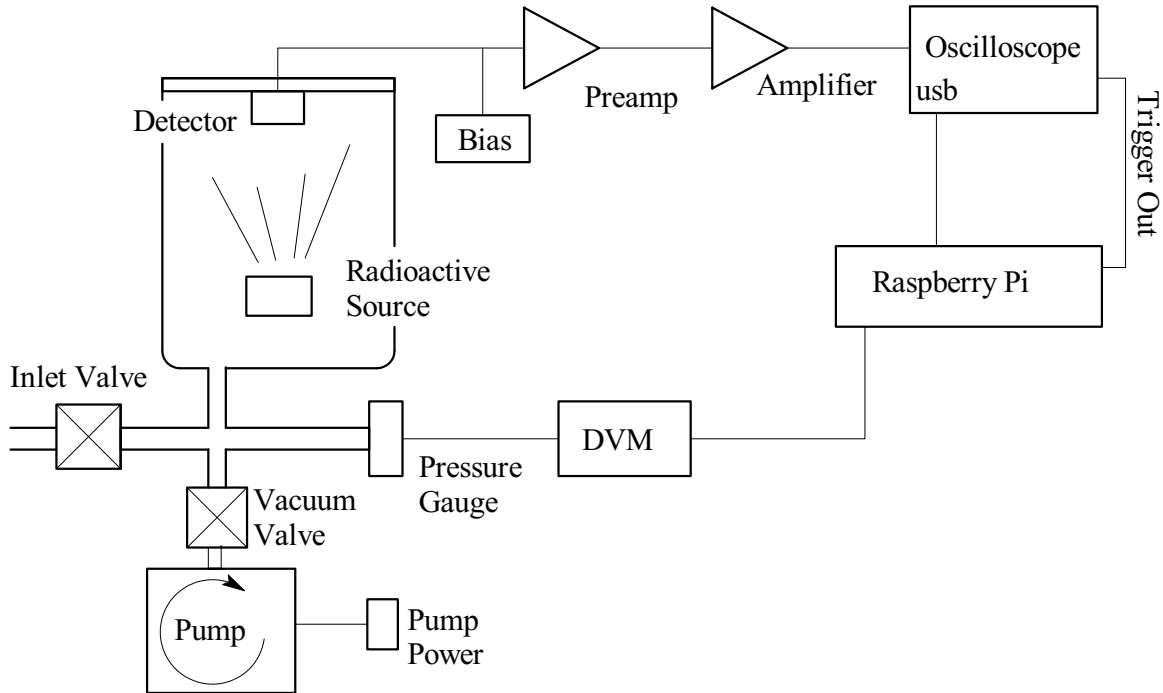


Figure 5: A Schematic diagram of our apparatus

accurate. Choose what to do based on what it is you are trying to measure.

There is another channel for getting data from the oscilloscope. The instrument puts a pulse out on its back panel every time it triggers. We have wired this pulse to a pin on the Raspberry Pi, and you can collect arrival times of these pulses. See the Appendix. Using this connection is a much more accurate and very much quicker way to count α particle detection rates than any method using scope queries.

As a first experiment, pump out the system and be sure the oscilloscope is healthily triggering on α particles.

Measure and plot the energy distribution by measuring quite a few trace amplitudes. 1,000 traces is a reasonable starting point.

Still at vacuum, measure the pulse arrival rate using the trigger monitor.

You will need the location and width of the energy distribution at vacuum and also the rate to make sense of any data you collect at other pressures.

If you want, devise a use of your data to test whether α particles arrive at random, uncorrelated with each other, as anticipated.

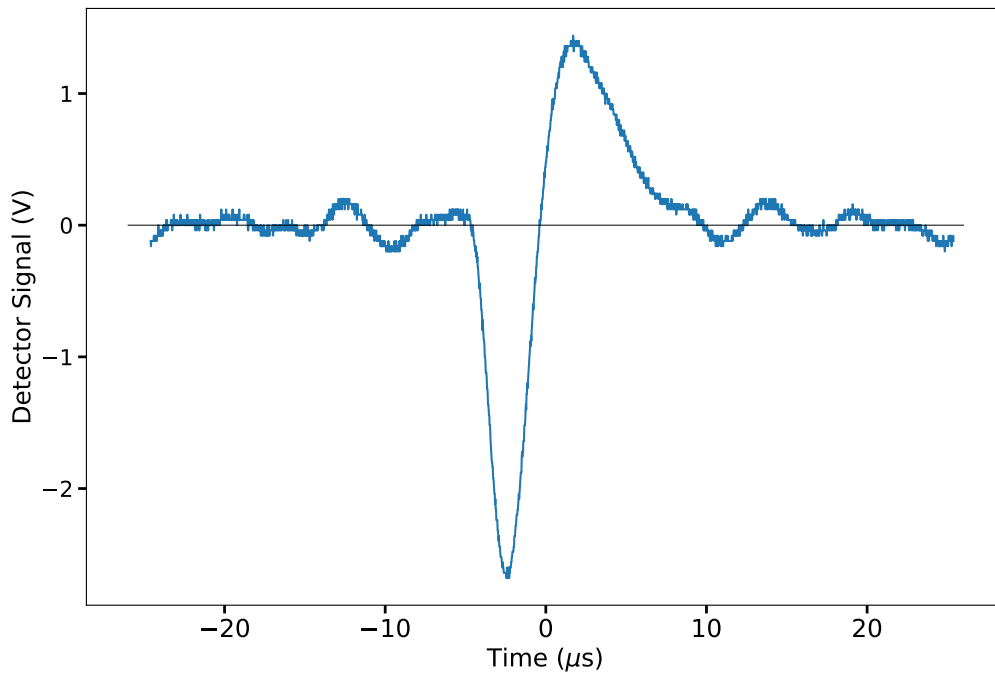


Figure 6: A single α particle detection. The system was near to vacuum and the energy of this particle is near to the original value before losses. The height of this shape is proportional to total particle energy. The oscilloscope which recorded this trace was set to trigger on a rising edge.

Once you have done this, you will be familiar enough with the apparatus and data collection procedures that you can design your own approach to exploring energy loss of α particles in air.

4 References

- Evans, **The Atomic Nucleus**, Pages 650 - 667.
- Melissinos, **Experiments in Modern Physics**, sections 5.2.2, 5.2.3, and 5.5.3 (first edition).

5 Appendix: Commands and Data Acquisition

We bought programmable digital multimeters. You can read them by running the command

```
ReadMM.py
```

which will return a DC voltage—currently attached to the pressure gauge. $7.2V = \text{vacuum}$. If you want to read something else, copy `ReadMM.py` to your own directory and modify it—but in this case, the DC voltage is the obvious choice.

We have modified `/home/public/ScopeRawQuery.py` to tolerate several USB oscilloscopes and other instruments. Please copy code from either `ScopeRawQuery.py` or `ReadMM.py` to whatever scripts you are using to collect data in order to correctly parse the USB instrument list.

There are two Alpha set ups. Look at computer/ experiment arrangement to find hostnames.

The original suite of programs includes

`ControlVacuum.py` Controls pressure in the experiment. Run the `-help` option first.

`TriggerPulseDetection.py` watches the pulses coming in on pin 5 from the trigger output of the scope. If that scope is set correctly to trigger on the alpha pulses, one can possibly estimate a pulse rate. Numbers are trigger time in microseconds. These numbers appear on your screen unless you redirect them to a file. (using `>`).

`ReadPressure.py` measures the DC voltage from the pressure sensor that is piped into channel 2 of the scope. It appears to (sometimes!) need quite some time to update. $7.2V$ corresponds to vacuum, low voltage to an atmosphere of pressure. Work near the vacuum end.

You can run these by typing their name, as though it is a command.

We have provided two programs whose code is in `/home/public/` to read data from the oscilloscope. They are not useful in their present form. Copy them to your folder, rename them and edit them.

`ScopeRawCapture.py` retrieves from the scope one or several traces. You can set which channel you collect from. You could insert commands to change scope settings as you like. Scope data transfer is slow, so we have put a sleep statement into the collection loop

`ScopeRawQuery.py` uses the measure facility of the oscilloscope to retrieve a value. You can change what you ask for. As above, edit this program to provide the data you actually want. Queries are faster than trace requests.

One important feature to notice: the commands to set a channel are different for retrieving a trace or a query. See the code.

See the programming manual for a full list of queries and other commands.

6 Appendix: Chamber Air Pressure Measurement

A solid state pressure transducer is used to convert pressure into voltage readings. The transducer uses strain gauges in a bridge configuration to measure the displacement produced in a solid plate by having an absolute vacuum (static) applied to one side and the working pressure applied to the other.

The transducer is calibrated at $P = 1$ atm and vacuum. Measure the atmospheric pressure when you do the lab using the Fortin barometer on the end of the bookshelves near the lab door. The other calibration point is obtained by pumping the chamber to $P \approx 0$ mmHg using the vacuum pump. A linear interpolation is used to determine the chamber pressure from the transducer response.

7 Appendix: The detector controls

BEFORE STARTING: Read the precautions on using the surface barrier detector located at the lab bench. Do not touch either the detector or the radioactive sample of ($\sim 1\mu\text{Ci}$ $^{241}\text{AmO}_2$).

1. Connect the apparatus as shown in Figure ??, and turn on the equipment. Open the UC530 application. Ensure that the bias voltage (“high voltage”) for the detector is set to zero. Check that when you turn it on it will be negative (hardware switch in one setup, software in the other).
2. Calibrate the pressure transducer as outlined above, and verify the separation between source and detector is ≈ 4.5 cm, recording the actual value. Evacuate the chamber when finished.
3. START THE NEGATIVE BIAS VOLTAGE AT 10-20 VOLTS AND TURN UP SLOWLY, MAKING SURE NOT TO EXCEED THE MAXIMUM ALLOWED FOR THE DETECTOR, THE BIAS POWER SUPPLY GOES UP TO 1000V!

Detector serial number	Max voltage
16-625J	-100 volts
16-721J	-75 volts
16-469A	-100 volts

4. Adjust the bias voltage for best signal to noise and adjust the amplifier gain. On one set-up with the older data acquisition system, adjust the gain so that the pulses are approximately 8 volts on the oscilloscope. With the newer data acquisition system you should set the gain so that the highest energy pulses are on scale.